MILITARY SPECIFICATION

SIGHT, BORE, REFLECTING: XM115E1

This specification is approved for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 This specification covers one type of instrument known as Sight, Bore, Reflecting: XM115 used for boresighting the Lightweight Company Mortar, 60MM,XM224. Throughout this specification this instrument is referenced as "Bore Sight."

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel Packaging and Packing of
MIL-I-45607	Inspection Equipment, Qcquisition, Maintenance, and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., Pa. 19137, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

DRAWINGS

US Army, Frankford Arsenal

F7573291	Collimator Projector
F11748094	Sight, Bore, Reflecting: XM115E1

PACKAGING DATA SHEET

11748094	Packaging of Sight, Bore, Reflecting: XM115E1

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Fabrication.- The boresight shall be manufactured in accordance with Drawing F11748094.

3.2 General Specifications.- The following requirements of MIL-F-13926 shall be met:

a. Order of precedence
b. Dimensions of tolerances
c. Inorganic protective surface finishes
d. Part identifications and marking
e. Workmanship

3.3 Lens Assembly, Reflecting Sight 11748099.- The Lens Assembly which is part of the boresight, shall have met the requirements of Dwg. 11748099 prior to assembly as a boresight.

3.4 Orientation.- The boresight shall be held securely in each of the two positions, shown in Figure 1, against the face of a simulated mortar tube. The face and axis of this tube shall be perpendicular to each other. The axis of this tube shall be inclined 45 degrees above the horizontal. The following conditions shall be met at each of the positions shown in Figure 1 unless otherwise specified.

3.5 Accuracy.

3.5.1 Collimation.- The line of sight of each lens assembly shall be parallel, within 1.0 mil, to the vertical plane through the tube axis of the mortar for Positions "A" and "B" of Figure 1. (See 3.5.2 and 3.5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

3.5.2 Cross level vials.- When the line of sight of each lens assembly is in each collimated position of 3.5.1 above, each cross level bubble shall be centered within 1/2 space width between two graduation lines.

3.5.3 Elevation level vial.- When the line of sight of the adjacent lens assembly, Position "A" of Figure 1, is in the collimated position of 3.5.1 above, the elevation level bubble shall be centered within 1/2 space width between two graduation lines.

3.6 Temperature.- The boresight shall be capable of withstanding temperature stabilization at +160°F and -50°F for a period of five hours at each extreme temperature. Subsequent to exposure to the temperatures specified herein, the boresight shall show no evedence of damage and shall meet the requirements of 3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassemby inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) boresights plus three (3) each of all items covered by SQAP. The three boresights shall be tested as specified herein for all the requirements of this specification. The items covered by SQAP shall be tested, as specified thereon, for all characteristics delineated on the SQAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Boresight.- Examination and tests related to Section 3 herein shall be performed on a class basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Collimation accuracy	3.5.1	4.6.2
102. Cross level vial accuracy	3.5.2	4.6.3
103. Elevation level vial accuracy	3.5.3	4.6.4

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One boresight shall be selected at random as a special sample from each 100 produced or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table II.

TABLE II - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawing visual
302.	General specification	3.2	MIL-F-13926 visual

NOTE: The tests in Table II shall be conducted at a temperature between +60° and +90°F.

4.4.2 Environmental.- Five boresights shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in Table I.

TABLE III - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Physical failure, post temperature cycle	3.6	4.6.5
304.	Collimation, post temperature cycle	3.6,3.5.1	4.6.5,4.6.2
305.	Cross level vial accuracy, post temperature cycle	3.6,3.5.2	4.6.5,4.6.3
306.	Elevation level vial accuracy, post temperature cycle	3.6,3.5.3	4.6.5,4.6.4

NOTE: The test for Characteristics 303 through 306 inclusive shall be conducted at a temperature between +60°F and +90°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved. (see 4.3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.1 Contractor design.- In the absence of Government inspection equipment design the contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Orientation.- The testing equipment utilized to mount and position the boresight for test shall consist of a simulated mortar tube, with base, as shown in Figure 2. The principle dimensions and surface finishes of the test fixture shall be as depicted in Figure 2.

4.5.2.3.2 Accuracy.- A target collimator conforming to Drawing F7573291, or equivalent, shall be utilized for the accuracy tests. The collimator shall be positioned in front of the simulated mortar tube so that it's optical-mechanical axis is aligned to the approximate center of the lens assembly of the boresight under test. The optical-mechanical axis of the collimator shall be horizontal within 20 seconds, and lay in the vertical plane that contains the center line of the simulated mortar within 0.125 inch. The vertical reticle line of the collimator shall be plumb and it's focus shall be adjusted for infinity.

4.5.2.3.3 Temperature testing.- The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F. The temperature gradient across the cross-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber in the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.6 Test methods and procedures.

4.6.1 Orientation.- The accuracy tests of 4.6.2 through 4.6.4 inclusive shall be performed with the boresight positioned on the special testing equipment that conforms to 4.5.2.3.1. A target collimator conforming to 4.5.2.3.2 shall be utilized with the simulated mortar.

4.6.2 Collimation accuracy.- Rotate the boresight to Position "A". Sight through the lens assembly and by minor rotational adjustments to the boresight plumb the reticle, as determined by the unaided eye. When the boresight is in this established Position "A" its line of sight shall meet the requirements of 3.5.1, as measured on the reticle of the target collimator. Rotate the boresight to Position "B" and again plumb the lens assembly reticle using the foregoing procedure. When the boresight is in the established Position "B" its line of sight shall meet the requirements of 3.5.1, as measured on the reticle of the target collimator.

4.6.3 Cross level vials accuracy.- Perform this test concurrent with the tests in 4.6.2. When the boresight is rotated to Positions "A" and "B", and the appropriate lens assembly meets the requirement of 3.5.1, the applicable cross level vial for each position shall meet the requirements of 3.5.2.

4.6.4 Elevation level vial accuracy.- Perform this test concurrent with the test for Position "A" in 4.6.2. When the borsight is rotated to Position "A", and its lens assembly meets the requirement of 3.5.1, the bubble in the elevation level vial shall meet the requirements of 3.5.3.

4.6.5 Temperature.- Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.3. The boresight shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature to -50°F. The boresight shall be maintained at this extreme temperature for a minimum of 5 hours. Upon completion of the 5 hour exposure the temperature of the chamber shall be raised, over a period of 8 hours plus or minus 15 minutes, to +160°F. The boresight shall remain at this temperature for a minimum of 5 hours. Upon completion of the 5 hour exposure the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The boresight shall then be removed from the test chamber and subjected to a visual, tactile, and audible, examination. The boresight shall show no evidence of damage. The boresight shall then be subjected to the tests in 4.6.2,4.6.3, and 4.6.4 to determine overall conformance to 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11748094. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The boresight is used to establish a definite line of sight when boresighting the Lightweight Company Mortar, 60MM, XM224.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable packaging data sheet (see 5.1).
c. Selection of applicable levels of preservation, packaging and packing.
d. Applicable stock number.
e. Provisions for First Article testing.

Custodian:

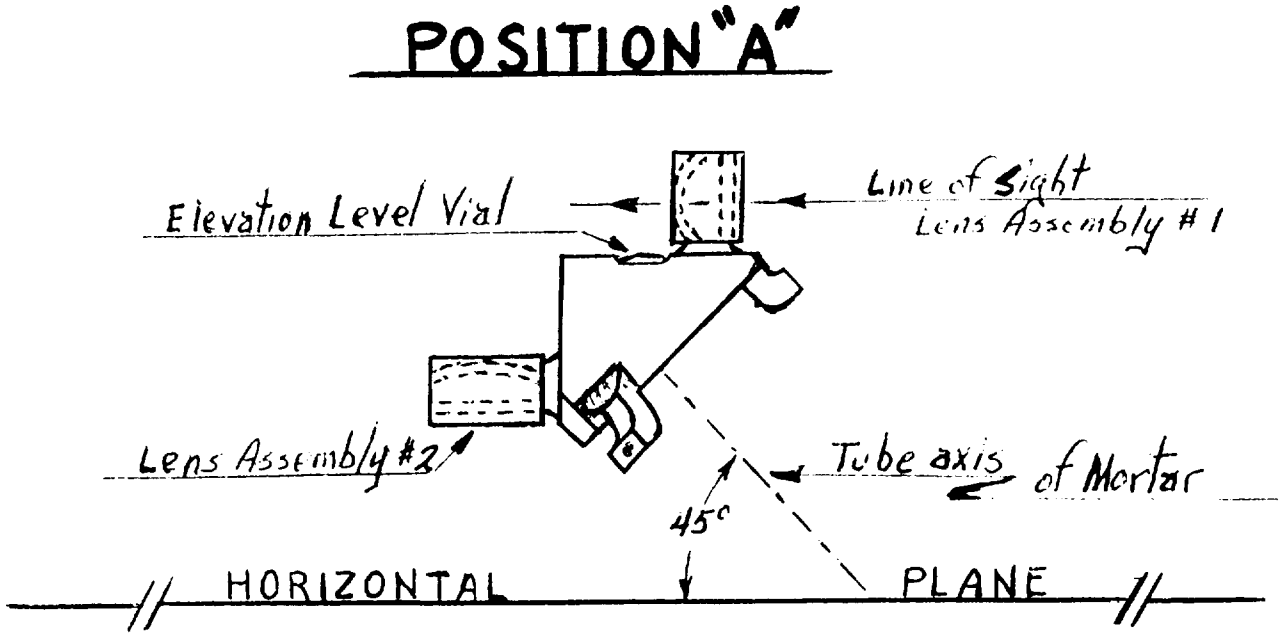
Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A685

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.



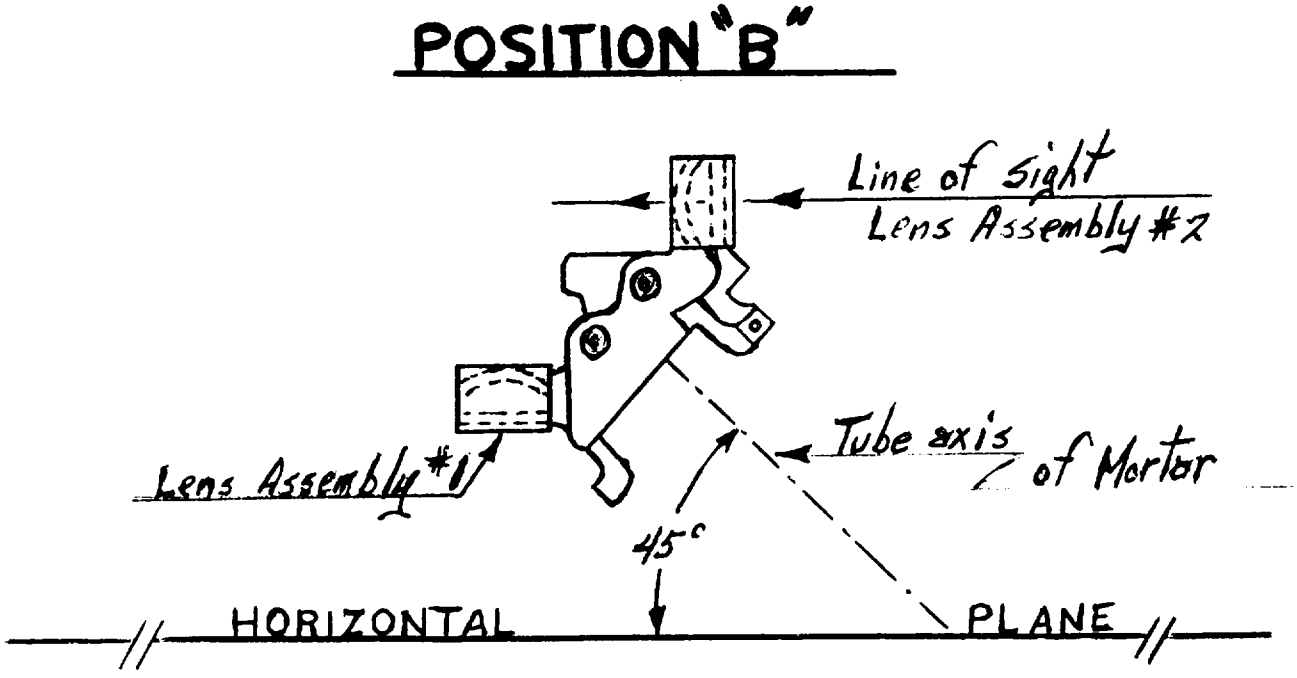


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

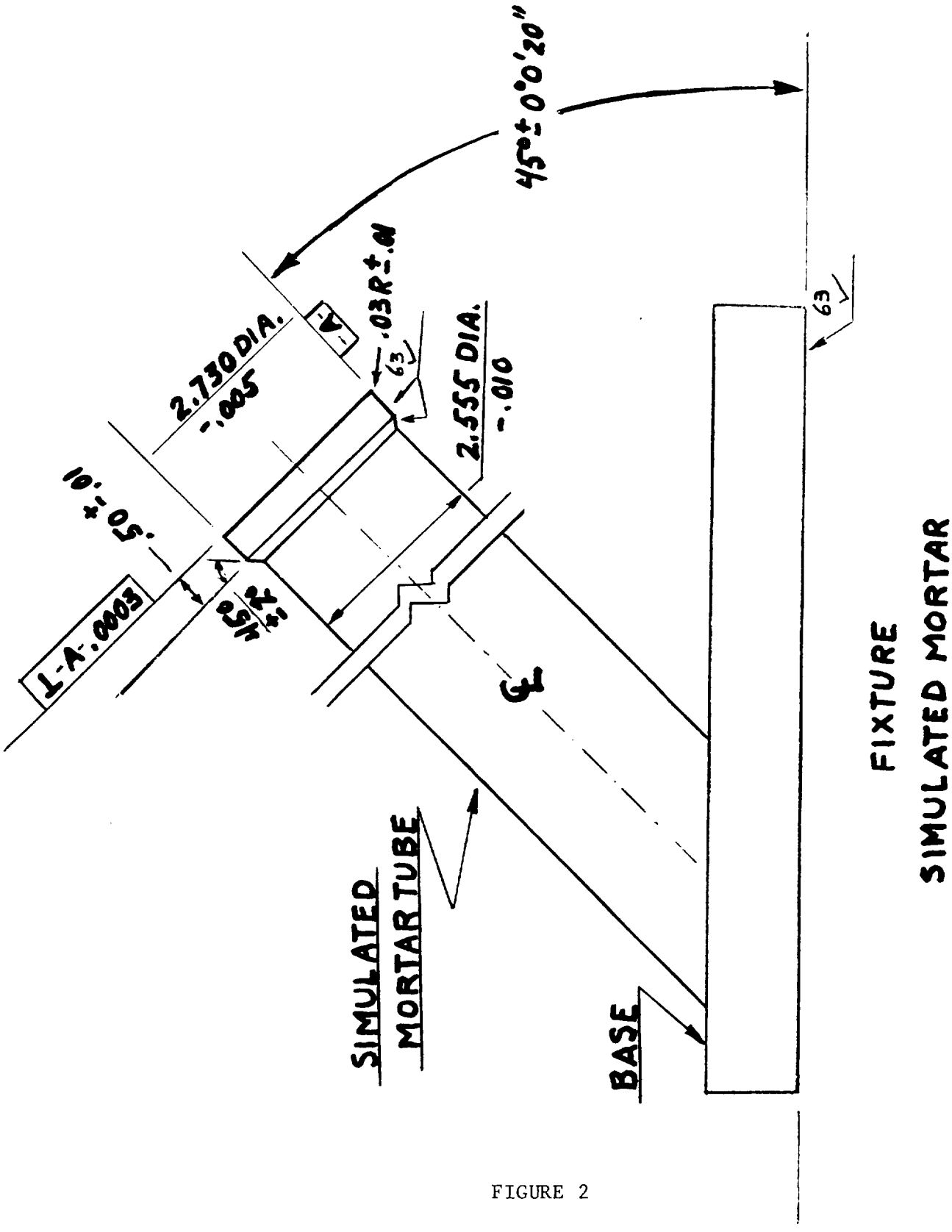


FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-48603(MU) Sight, Bore, Reflecting: XM115E1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Phila., PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.